MODULE NAME:	MODULE CODE:
OPEN SOURCE CODING (INTERMEDIATE)	OPSC6312/p/w

ASSESSMENT TYPE: POE (PAPER AND MARKING RUBRIC)

TOTAL MARK ALLOCATION: 300 MARKS

TOTAL HOURS: A MINIMUM OF 35 HOURS IS SUGGESTED TO COMPLETE THIS ASSESSMENT

By submitting this PoE, you acknowledge that you have read and understood all the rules as per the terms in the registration contract, in particular the assignment and assessment rules in The IIE Assessment Strategy and Policy (IIE009), the intellectual integrity and plagiarism rules in the Intellectual Integrity Policy (IIE023), as well as any rules and regulations published in the student portal.

INSTRUCTIONS:

1. ***No material may be copied from original sources, even if referenced correctly, unless it is a direct quote indicated with quotation marks. No more than 10% of the assignment may consist of direct quotes.***
2. ***Please ensure that you submit your assignment through Turnitin. Please make sure you attach a similarity report to your POE if you are required to submit a hard copy of your PoE.***
3. ***Make a backup of your assignment before handing it in.***
4. *Follow all instructions on the PoE cover sheet.*
5. *This is a group assignment.*

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Referencing Rubric

Providing evidence based on valid and referenced academic sources is a fundamental educational principle and the cornerstone of high-quality academic work. Part of achieving this quality is referencing in a way that is consistent and congruent with the requirements of the referencing style being used.

Therefore, inconsistent and/or incongruent referencing will result in a penalty of **a maximum of ten percent being deducted from the overall percentage** awarded to your assessment submission.

Please note that **evidence of plagiarism** in the form of copied or unreferenced work, absent reference lists, or exceptionally poor referencing **may result in action being taken in accordance with The IIE's Intellectual Integrity and Property Rights Policy (IIE023)**. Similarly, **evidence of excessive AI usage may result in action being taken in accordance with The IIE's Student Conduct, Discipline and Safety Policy (IIE015)**.

Markers are required to provide feedback to students by **circling/underlining the information in the table below that best describes the student's work and by adding constructive commentary where appropriate.** The examples provided are not exhaustive but illustrate the errors.

Deductions

- Where the student's work contains **five or more errors** aligned to the **minor errors column** below, **deduct 5% from the overall percentage.**
- Where the student's work contains **five or more errors** aligned to the **major errors column** below, **deduct 10% from the overall percentage.**
- **Where both minor and major errors** (e.g. two minor and three major, etc.) are present, **deduct 10% only** (and not 5% or 15%) from the overall percentage.

Required: **Consistent and congruent referencing**	**Minor errors** **Deduct 5% from overall percentage.** **Example: if the response receives 70%, deduct 5%. The final mark is 65%.**	**Major errors** **Deduct 10% from the overall percentage.** **Example: if the response receives 70%, deduct 10%. The final mark is 60%.**
Consistency • The **correct referencing style** for the **discipline** – i.e., either **Harvard**, OR **APA** (for Psychology)**,** OR **Law**, OR **IEEE** (for ICT/Engineering) – has been used consistently for all in-text references and in the bibliography/reference list. • **Concepts** and **ideas** that are **quoted** and/or **paraphrased** are referenced consistently throughout. • **Position of the in-text reference**: an in-text reference is positioned consistently where appropriate for every quote and paraphrase.	**Minor inconsistencies:** • The referencing style used is generally consistent with what is required, but there are one or two changes/errors in the format of in-text referencing and/or in the bibliography/reference list. • For example, page numbers for direct quotes in-text have been provided for one source, but not in another. Or, two book chapters in the bibliography/reference list have been referenced in two different formats. Or, the publication year has been placed after the author name in one bibliography/reference list entry, and after the source title in another, etc. • Concepts and ideas in quotes and/or paraphrases are typically referenced, but a full in-text reference is missing or incomplete from one or two small sections of the work. • Position of the references: in-text references are only given at the beginning and/or end of every paragraph.	**Major inconsistencies:** • Poor and wholly inconsistent referencing style used in-text and/or in the bibliography/reference list. • Multiple referencing styles for the same source types have been used. • For example, the format for direct quotes in-text and/or book chapters in the bibliography/reference list and/or year of publication in the bibliography/reference list is different across multiple instances. • Concepts and ideas in quotes and/or paraphrases are haphazardly referenced in-text. • Position of the references: in-text references are only given at the beginning or end of large sections of work.
Feedback on referencing consistency:		
Congruency • **Each source** reflected within **in-text references** is included accurately in the **bibliography/reference list.** • All **bibliography/reference list** entries are in the **required order** for the referencing style used (e.g. alphabetical, alphabetical under subheadings, numerical). • All **direct quotes** and **paraphrases** have been **integrated appropriately** into the text using introductory phrases, accurate grammar, etc.	**Minor incongruences:** • There is largely a match between the sources presented in-text and those in the bibliography/reference list, but one or two sources that appear in-text do not appear in the bibliography/reference list, or vice versa. Or key source information is missing from one or two in-text references or bibliography/reference list entries only (e.g. publication year, city of publication, URL date accessed, etc.). • There is a clear and largely accurate ordering of sources in the bibliography/reference list as required by the referencing style used, but with one or two references out of order. • An attempt has been made for source integration into the text using appropriate introductory phrases and grammar, but one or two quotes or paraphrases do not flow as clearly or logically within the sentence structure as they could.	**Major incongruences:** • No relationship/several incongruencies between the in-text referencing and the bibliography/reference list. • For example, multiple sources are included in-text, but not in the bibliography, and/or vice versa. Key source information is missing from multiple in-text references and/or reference list entries. A URL link, rather than the actual reference, is provided in the bibliography. Sources are repeated in the reference list, etc. • Most sources are listed in a haphazard order throughout the bibliography/reference list. • Few to no appropriate introductory phrases or rules of grammar have been applied, and many direct quotes and/or paraphrases feel disconnected from the flow of the text.
Feedback on referencing congruency:		
Overall feedback on referencing, with suggested improvements:		

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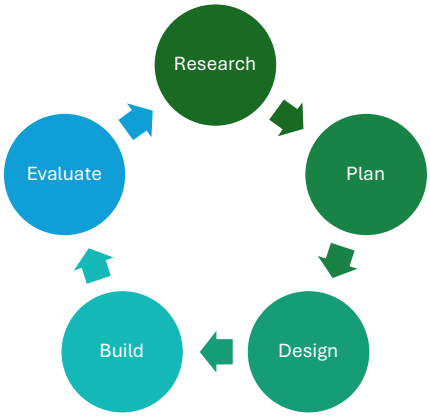
Background

This portfolio of evidence comprises two parts submitted during the semester, and the finalisation of your PoE, each of which takes you through conceptualising, designing, and building a mobile application.

You are going to create an app of **your choice** for a change. The app must make use of an **API** that you find online, or you are welcome to create it yourself, utilising the knowledge learned in PROG6212 and APPR6312. The app must include the stated features described in the introduction section below.

Finally, the app will need to be prepared so that it can be published on the Google Play Store, as well as making use of unit testing and GitHub Actions for running the necessary tests.

To be able to create the app, you will need to use all the techniques you learned during this course, as well as do your research on how to complete some of the tasks required of you. It is also critical that you approach the project in a methodical and planned manner. For this reason, the project will follow a Research – Plan – Design – Build – Evaluate cycle:



This PoE will run for one full cycle. After one full cycle, you can always restart the process to further refine your app.

The different phases of the cycle will be related to the different tasks as follows:

Part 1: Research, Planning and Design

Part 2: Build

Final PoE: Build and Evaluate

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Instructions

Each of the following parts is built on top of the other. Be sure to complete or update one part before moving on to the next. Each part will be individually assessed by your lecturer, and feedback for each part will be provided.

The exact features, design, and layout of your app are up to you, but each app must be able to complete at least the following features:

- The user must be able to **register** and **log in** to the app. Make sure to encrypt the password
- Make use of single sign-on (**SSO)** **(PoE only)**
- The user must be able to change their **settings** in the app.
- Connect to a **REST API** you create that is connected to a database or use any storage mechanism that fits the idea you have, that exists on the internet.
- **Offline mode** with Sync: Create a feature that allows users to perform offline actions with synchronisation capabilities once they reconnect. This does not need to be the whole app, but at least some features. Can make use of RoomDB or SQL Lite **(PoE only)**
- Implementation of **Real-time Notification**: Implement a push notification system for real-time updates and alerts **(PoE only)**
- **Multi-language** support that includes at least two South African Languages **(PoE only)**

These features are the **minimum** that is required. Your app also needs to work **logically** and be **easy to use**. Speak to your lecturer about how your app can best implement all these features.

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Part 1 — Research, Planning and Design (Marks: 100)

Learning Unit 1

At the end of this specific part, students should be able to:

- Explain how to connect to a RESTful Application Programming Interface (API).
- Define the features that the app will include.
- Define the use case of their app.

This part is composed of two sections, weighted equally:

- Research
- Planning and design

Research (Marks: 50)

For this part, you are required to research **three existing apps** that are like the one you want to create and are available on the Android operating system. You will need to present your findings in the form of a **report (1000 to 1500 words)** with the following sections:

- Introduction
- Research on each app, including:
  - Overview of the app
  - Strengths and weaknesses of the app
  - How you think the app was implemented based on your knowledge of Android Studio app development
  - Screenshots of the app
- A comparison of all three apps
- A list of the best features of all the apps that you want to use in your final app
- Conclusion
- References

The information **must be submitted in the form of a report**. Make sure you **reference** this part correctly.

- Make sure that the size of the PDF file does not exceed 10 MB.

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- Please note that if you don't find an app that matches exactly, then look for one that matches at least some of the features.

Planning and Design (Marks: 50)

For this part, you will need to design your app fully. The purpose of this part is to ensure you know exactly what you need to build and how you will build it before you start with Part 2. You will need to present your design as a typed PDF document (1000 to 1500 words) with the following sections:

- Introduction.
- A brief overview of the app, including a name for the app, an initial icon design, and a description of the innovative features that you are planning to include.
- A detailed list of the requirements for the app.
  The requirements must **include all the minimum requirements from this document**, but you must elaborate on each of those requirements **rather than just copying them.** Remember to add the features that you want to include, for example, gamification. The requirements must be **detailed enough** for someone else to be able to build your app by just reading this document! You need to make these clear since you will be marked on these for Part 2 and the PoE.
- **User interface design**, including a **mockup for each screen**, a **description of the purpose of the screen**, and a **diagram showing how the user navigates between the various screens**.
- Design what your API will do for your application and what data it will send and receive. Also, state how you are going to create your API and what hosting you will be using. If you have found an existing API you would use, please state this here and how best to use it. You will also need to state what database you would use with it, for example, Firebase.
- UML diagrams that include how the API is connected to other systems in the application.
- Detailed listing of the data that the app needs to capture from the user and store, including data types.
- Project plan detailing deadlines and milestones for the project in the format of a Gantt chart. Break down the tasks into smaller tasks than just "implement the prototype". Include some time for testing and bug fixing.
- Conclusion.
- References.

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Submit the following for Part 1:

- **Research Report** and **Planning and Design Document** in PDF format.
- If applicable, **a short write-up** (maximum 500 words) on the manner in which **AI tools were used and cited** during the completion of this assessment. Examples could include code snippets, debugging errors, image generation, etc.

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Part 2 — App Prototype Development (Marks: 100)

Learning Units 1 and 2

At the end of this specific part, students should be able to:

- *Use a RESTful API in an Android app.*
- *Use an external library to solve a programming problem.*
- *Connect an app to an appropriate SDK.*
- Conduct detailed unit testing.

For this part, you will need to build a fully working prototype. This prototype needs to include all the features listed in the instructions section of this document **(except for the requirements marked for the PoE),** based on your own design and user interface layout. The app must compile and run for marks to be awarded for the app's features.

Some of the features specified in your design document can be deferred until the final PoE submission, but you need to include at least a few of them for Part 2. And final assets (for example, images) do not need to be included in the prototype.

You must create a demonstration video showing the prototype running on your mobile phone. For the video, make sure that you show the app performing the following functions:

- The user must be able to register and log in to the app, and encrypt the password of the user.
- The user must be able to change their settings in the app.
- Connect to a REST API you create that is connected to a database, or use any that fits your idea that exists on the internet. Remember, this will need to be hosted to connect to your application. You may use anything for this.
- Your features that were specified in Part 1. Look at the rubric to see the weighting.

Include in the demonstration video that will effectively demonstrate all the features of the app and be fully professional.

The video must include a voice-over explaining what you are showing. The video must also show what data is stored in the online-hosted authentication service, API, and database.

The app must have a user-friendly and appropriate user interface that can handle invalid inputs made by the user without crashing. The app must work with only minor bugs and errors.

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Version Control with GitHub: Initialise the repository with a **README file**. **Commit** and **push** your project files to the **GitHub repository**. Regularly commit and push your code as you make progress.

Conduct automated testing on the main functionality of your app. Make use of **GitHub Actions** to run tests and build your code to make sure it will work on not just your computer. Use this guide, which will assist in creating this:

- https://github.com/marketplace/actions/automated-build-android-app-with-github-action [Accessed on 03 November 2025].
- *https://github.com/IMAD5112/Github-actions/blob/main/.github/workflows/build.yml* [Accessed on 03 November 2025].

Documentation:

Prepare a comprehensive report detailing the purpose of the app, its design considerations, and the utilisation of GitHub and GitHub Actions. Include all of this inside your **README file on GitHub**. Add images and make the readme file your own.

Submission:

For your submission, present the following:

- The complete Kotlin source code for the Android app must be submitted on GitHub. ***No zip files are allowed.***
- Add comments to your code, and don't forget to reference them.
- Make use of logging to show your understanding of your code.
- Your README file for your app that is also on GitHub.
- A video presentation showcasing all the app's features. Include a link to this video in the README file.
- If applicable, **a short write-up** (maximum 500 words) on the manner in which **AI tools were used and cited** during the completion of this assessment. Examples could include code snippets, debugging errors, image generation, etc.

You can upload your video to YouTube, make it unlisted, and add that link to your README file. You can also use any video recording software you like. Just make sure to include a voice-over to show your understanding.

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Part 3 — Final Asset Development, Additions, Tweaking and Final Submission (Marks: 100)

Learning Units 1 to 4

At the end of this specific part, students should be able to:

- *Use blob storage.*
- *Use a NoSQL database to store data.*
- *Use a NoSQL database to read data.*
- *Prepare an app for publication in the Play Store.*

The final app needs to run on a mobile phone and not an emulator. This means that you need to plan well in advance to record the required video on a mobile phone if you don't own an Android phone yourself.

For this submission, the app should be a fully working app without errors that has been updated based on the feedback your lecturer has provided you throughout the course. The app must **compile** and **run** for marks to be awarded for the app's features.

You must create a demonstration video showing that the application is working. For the video, make sure that you show the app performing the following functions:

- The user must be able to **register** and **log in** to the app. Make sure to encrypt the password
- Make use of single sign-on (**SSO) (PoE only).**
- The user must be able to change their **settings** in the app.
- Connect to a **REST API** you create that is connected to a database, or use any that fits your idea that exists on the internet.
- **Offline mode** with Sync: Create a feature that allows users to perform offline actions with synchronisation capabilities once they reconnect. This does not need to be the whole app, but at least some features. Can make use of RoomDB or SQLite **(PoE only).**
- Implementation of **Real-time Notification**: Implement a push notification system for real-time updates and alerts **(PoE only).**
- **Multi-language** support that includes at least two South African languages **(PoE only).**
- Your features that were specified in Part 1. Look at the rubric to see the weighting.
- An app icon and final image assets.

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You may consult any resources of your choice to implement push **notifications**. These resources may assist you:

- Push Notifications: https://firebase.google.com/docs/cloud-messaging [Accessed on 03 November 2025].

Include in the demonstration video that will effectively demonstrate all the features of the app and be fully professional. The video must include a voice-over explaining what you are showing. The video must also show what data is stored in the online-hosted authentication service, API, and database.

The app must have a user-friendly and appropriate user interface that can handle invalid inputs made by the user without crashing. The app must be working.

Version Control with GitHub: Initialise the repository with a **README file**. **Commit** and **push** your project files to the **GitHub repository**. Regularly commit and push your code as you make progress. Since this is the final PoE, make sure to tag your final commit as the ***final PoE*****.**

Conduct automated testing on the main functionality of your app. Make use of **GitHub Actions** to run tests and build your code to make sure it will work not on just your computer. Use this guide, which will assist in creating this:

- https://github.com/marketplace/actions/automated-build-android-app-with-github-action [Accessed on 03 November 2025].
- *https://github.com/IMAD5112/Github-actions/blob/main/.github/workflows/build.yml* [Accessed on 03 November 2025].

Documentation:

Prepare a comprehensive report detailing the purpose of the app, its design considerations, and the utilisation of GitHub and GitHub Actions. Include all of this inside your **README file on GitHub**. Add images and make the README file your own. Release notes need to be added as well, which list all the updates that have been made to the app since the prototype. The release notes must clearly explain which of your innovative features were added to this version.

Publishing:

The app will also need to be ready to be published on the Google Play Store.

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Submission:

For your submission, present the following:

- The complete Kotlin source code for the Android app must be submitted on GitHub. ***No zip files are allowed.***
- Add comments to your code, and don't forget to reference them.
- Make use of logging to show your understanding of your code.
- Your README file for your app that is also on GitHub.
- A video presentation showcasing all the app's features. Include a link to this video in the README file.
- Release notes that list all the updates that have been made to the app since the prototype. The release notes must clearly explain which of your innovative features were added to this version.
- Evidence of preparation for publication: signed APK, screenshots, and a screenshot of the app uploaded on the Google Play Console and/or published on the Google Play Store, if applicable.
- If applicable, **a short write-up** (maximum 500 words) on the manner in which **AI tools were used and cited** during the completion of this assessment. Examples could include code snippets, debugging errors, image generation, etc.

You can upload your video to YouTube, make it unlisted, and add that link to your README file. You can also use any video recording software you like. Just make sure to include a voice-over to show your understanding.

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Appendix A - PoE Marking Rubrics

Markers – Please note that the rubrics below must be used to evaluate the students' responses to the relevant assignment questions. Please clearly indicate the specific mark you allocate for each rubric criterion to show how you reached the question total. Also, please provide constructive feedback to ensure students and moderators can follow your marking logic based on the rubric criteria. The most important point is that many markers across different campuses will be marking. The rubric needs to promote the validity and reliability of their marking practices. In addition, there is a separate memorandum that provides additional marking guidance – please ensure you get this from your relevant campus administrator.

While reading the student's submission, the marker should consider the goals of the task and the description of each qualitative level. The marker should match the student's work to the appropriate qualitative level, highlighting areas of achievement or areas that were not adequately addressed. Once the marker has considered all of this, then only should the student be assigned a mark.

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Assessment Sheet **(Marking Rubric)**

Please note: Tear off this section and **attach** it to your work when you submit it/ If this is an online submission, then this information needs to be included in the online submission.

MODULE NAME:	MODULE CODE:
OPEN SOURCE CODING (INTERMEDIATE)	OPSC6312

STUDENT NAME:
STUDENT NUMBER:

PART 1 – Research					
Marking Criteria	**Does not meet the required standard**	**Meets the required standard**	**Partially exceeds the required standard**	**Greatly exceeds the required standard.**	**Feedback**
Research: Introduction **[5 Marks]**	• No introduction included. • The introduction does not link to the rest of the document.	• Acceptable and links to the rest of the document, but does not explain the purpose of the research.	• Acceptable and links to the rest of the document with some details about the purpose of the research.	• An excellent introduction that explains the purpose of the research and the sections included in the document.	
	0 – 1 Marks	**2 – 3 Marks**	**4 Marks**	**5 Marks**	
Research: App 1 research **[8 Marks]**	• Not included or sections missing. • All sections are included, but limited details are provided.	• All sections are included, but more details could be added to most of the sections.	• All sections are included, but more details could be added to some of the sections.	• Comprehensive discussion of the strengths and weaknesses, and motivating why features are considered innovative.	
	0 – 1 Marks	**4 – 5 Marks**	**6 Marks**	**7 – 8 Marks**	

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PART 1 – Research					
Marking Criteria	**Does not meet the required standard**	**Meets the required standard**	**Partially exceeds the required standard**	**Greatly exceeds the required standard.**	**Feedback**
Research: App 2 research **[8 Marks]**	• Not included or sections missing. • All sections are included, but limited details are provided.	• All sections are included, but more details could be added to most of the sections.	• All sections are included, but more details could be added to some of the sections.	• Comprehensive discussion differentiating between the strengths and weaknesses and motivating why features are considered innovative.	
	0 – 1 Marks	**4 – 5 Marks**	**6 Marks**	**7 – 8 Marks**	
Research: App 3 research **[8 Marks]**	• Not included or sections are missing. • All sections are included, but limited details are provided.	• All sections are included, but more details could be added to most of the sections.	• All sections are included, but more details could be added to some of the sections.	• Comprehensive discussion differentiating between the strengths and weaknesses and motivating why features are considered innovative.	
	0 – 1 Marks	**4– 5 Marks**	**6 Marks**	**7-8 Marks**	
Research: Comparison **[10 Marks]**	• No comparison is included, nor is the comparison in a visual format. • Comparison in a visual format, but only includes either differences or similarities.	• Comparison in a visual format with a good number of similarities and differences, but the presentation could be improved.	• Comparison in a visual format with a good number of similarities and differences.	• Comprehensive comparison that shows all the differences and similarities at a glance.	
	0 – 4 Marks	**5 Marks**	**6 – 7 Marks**	**8 – 10 Marks**	

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PART 1 – Research					
Marking Criteria	**Does not meet the required standard**	**Meets the required standard**	**Partially exceeds the required standard**	**Greatly exceeds the required standard.**	**Feedback**
Research: List of features to include **[6 Marks]**	• No list of features included. • A very short list included with no motivations.	• A good list of features is included, but with little or no motivation.	• A good list of features is included with some motivation.	• An excellent list of features with motivations included for why these features are desirable.	
	0 – 2 Marks	**3 – 4 Marks**	**5 Marks**	**6 Marks**	
Research: Conclusion **[5 Marks]**	• No conclusion included. Conclusion does not link to the rest of the document.	• Acceptable and links to the rest of the document.	• Acceptable and links to the rest of the document. The link to the design is mentioned but not elaborated on.	• An excellent conclusion that links to the document and the design.	
	0 – 1 Marks	**2 – 3 Marks**	**4 Marks**	**5 Marks**	
Planning and Design: Introduction and Conclusion **[5 Marks]**	• No introduction or conclusion included. • Either the introduction or the conclusion is missing.	• A good introduction and conclusion are included, as well as links to the rest of the document.	• A good introduction and conclusion are included, as well as links to the rest of the document. • Mention is made of the research document.	• An excellent introduction and conclusion that links to the research document as well as the content of the design document.	
	0 – 1 Marks	**2 – 3 Marks**	**4 Marks**	**5 Marks**	

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PART 1 – Planning and Design					
Marking Criteria	**Does not meet the required standard**	**Meets the required standard**	**Partially exceeds the required standard**	**Greatly exceeds the required standard.**	**Feedback**
Planning and Design: Overview of the app **[10 Marks]**	• No overview is included, and no innovative features are mentioned. • The overview needs more details.	• A good overview is included with some innovative features.	• A good overview is included with some innovative features and an attempt to link to the rest of the document.	• An excellent overview that sets the stage for the rest of the document.	
	0 – 4 Marks	**5 Marks**	**6 – 7 Marks**	**8 – 10 Marks**	
Planning and Design: Detailed list of requirements **[10 Marks]**	• No requirements are included, and the required features are missing. • Only required features are included with some details, but no additional features from the research are mentioned.	• Required features and the students' requirements are included, but more details are needed in places.	• Required features and the students' requirements are included, with more detail needed only in a few places.	• An excellently detailed list of features that describes all the features, including the student's own research in detail.	
	0 – 4 Marks	**5 Marks**	**6 – 7 Marks**	**8 – 10 Marks**	
Planning and Design: Creation of the REST API **[10 Marks]**	• Little to no understanding of REST API.	• Some understanding of the implementation of making use of the API with the flow on the UML is somewhat correct.	• Only UML created or detailed understanding of making use of a REST API to complete the application.	• Good, detailed understanding of how to make use of the REST API to complete the application. • UML Diagram created.	
	0 – 4 Marks	**5 Marks**	**6 – 7 Marks**	**8 – 10 Marks**	

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PART 1 – Planning and Design					
Marking Criteria	**Does not meet the required standard**	**Meets the required standard**	**Partially exceeds the required standard**	**Greatly exceeds the required standard.**	**Feedback**
Planning and Design: User interface design **[10 Marks]**	• No design is included, and only a very brief description with no mockups is provided. • Only mockups or reasonably detailed descriptions included, not both.	• Mockups and descriptions included, but no diagrams showing navigation.	• Mockups and descriptions included, and navigation diagrams included with some missing details.	• Excellent mockups, together with descriptions and a diagram explaining the navigation.	
	0 – 4 Marks	**5 Marks**	**6 – 7 Marks**	**8 – 10 Marks**	
Planning and Design: Project Plan **[5 Marks]**	• No project plan included. • Project plan is included, but it is not logical.	• Logical with some errors.	• Logical with only minor errors.	• Logical and fully detailed with no errors.	
	0 – 1 Marks	**2 – 3 Marks**	**4 Marks**	**5 Marks**	**Part 1 Total /100**

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PART 2 – App Prototype Development					
Marking Criteria	**Does not meet the required standard**	**Meets the required standard**	**Partially exceeds the required standard**	**Greatly exceeds the required standard**	**Feedback**
The app runs on a mobile device. **[5 Marks]**	• The app doesn't run at all. • The app runs, but lots of crashes happen.	• The app runs with one or two errors or crashes. • Runs on the emulator	• The app runs with very infrequent errors or crashes.	• The app runs without errors.	
	0 – 1 Marks	**2 – 3 Marks**	**4 Marks**	**5 Marks**	
Feature: **sign in** **[10 Marks]**	• Feature not included. • Feature implemented but very buggy.	• Feature is working mostly with some bugs.	• Feature is working with only minor bugs.	• Feature excellently implemented.	
	0 – 4 Marks	**5 Marks**	**6 – 7 Marks**	**8 – 10 Marks**	
Feature: Settings menu that makes sense for the application **[10 Marks]**	• Feature not included. • Feature implemented but very buggy.	• Feature is working mostly with some bugs.	• Feature is working with only minor bugs.	• Feature excellently implemented.	
	0 – 4 Marks	**5 Marks**	**6 – 7 Marks**	**8 – 10 Marks**	
Creation/use of the REST API **[10 Marks]**	• API not included. • API was created, but very buggy.	• API is working mostly with some bugs.	• API is working with only minor bugs.	• API excellently created/used	
	0 – 4 Marks	**5 Marks**	**6 – 7 Marks**	**8 – 10 Marks**	

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PART 2 – App Prototype Development					
Marking Criteria	**Does not meet the required standard**	**Meets the required standard**	**Partially exceeds the required standard**	**Greatly exceeds the required standard**	**Feedback**
Integration of your REST API in your app	• Feature not included. • Feature implemented but very buggy.	• Feature is working mostly with some bugs.	• Feature is working with only minor bugs.	• Feature is excellently integrated.	
[10 Marks]	**0 – 4 Marks**	**5 Marks**	**6 – 7 Marks**	**8 – 10 Marks**	
Feature: User Defined 1	• Feature not included. • Feature implemented but very buggy.	• Feature is working mostly with some bugs.	• Feature is working with only minor bugs.	• Feature excellently implemented.	
[10 Marks]	**0 – 4 Marks**	**5 Marks**	**6 – 7 Marks**	**8 – 10 Marks**	
Feature: User Defined 2	• Feature not included. • Feature implemented but very buggy.	• Feature is working mostly with some bugs.	• Feature is working with only minor bugs.	• Feature excellently implemented.	
[10 Marks]	**0 – 4 Marks**	**5 Marks**	**6 – 7 Marks**	**8 – 10 Marks**	
Feature: User Defined 3	• Feature not included. • Feature implemented but very buggy.	• Feature is working mostly with some bugs.	• Feature is working with only minor bugs.	• Feature excellently implemented.	
[10 Marks]	**0 – 4 Marks**	**5 Marks**	**6 – 7 Marks**	**8 – 10 Marks**	

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PART 2 – App Prototype Development					
Marking Criteria	**Does not meet the required standard**	**Meets the required standard**	**Partially exceeds the required standard**	**Greatly exceeds the required standard**	**Feedback**
User interface **[10 Marks]**	• Not user-friendly, confusing, and illogical. • Somewhat user-friendly with lots of mistakes in the design.	• Mostly user-friendly, with some mistakes in the design. • Somewhat consistent use of layout, fonts, and colours.	• Mostly user-friendly, with some attempt made at making the capture process easier for the user. • Mostly consistent use of layout, fonts, and colours.	• Excellent and user-friendly design. • Completely consistent use of layout, fonts, and colours.	
	0 – 4 Marks	**5 Marks**	**6 – 7 Marks**	**8 – 10 Marks**	
Github, Readme and automated testing **[10 Marks]**	• Zip file added or file upload used for GitHub.	• Multiple commits and unit testing are not working. • No readme file.	• Automated testing includes, but is not working correctly. • Somewhat readme file created.	• Readme file created and implemented correctly. • Automated testing is in place. • Multiple commits.	
	0 – 4 Marks	**5 Marks**	**6 – 7 Marks**	**8 – 10 Marks**	
Demonstration video **[5 Marks]**	• No demonstration video included. • Informal and unprepared, or not showing all the features or no voice-over included.	• It is not completely professional, but all features are demonstrated.	• Mostly professional with some issues, but all features were demonstrated.	• Fully professional video showing all the required features in detail.	
	0 – 1 Marks	**2 – 3 Marks**	**4 Marks**	**5 Marks**	**Part 2 Total /100**

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PORTFOLIO OF EVIDENCE (POE)					
Marking Criteria	**Does not meet the required standard**	**Meets the required standard**	**Partially exceeds the required standard**	**Greatly exceeds the required standard**	**Feedback**
The app runs on a mobile device. **[5 Marks]**	• The app doesn't run at all. • The app runs, but lots of crashes happen.	• The app runs with one or two errors or crashes. • Runs on the emulator	• The app runs with very infrequent errors or crashes.	• The app runs without errors.	
	0 – 1 Marks	**2 – 3 Marks**	**4 Marks**	**5 Marks**	
Feature: SSO login **[10 Marks]**	• Feature not included. • Feature implemented but very buggy.	• Feature is working mostly with some bugs.	• Feature is working with only minor bugs.	• Feature excellently implemented.	
	0 – 4 Marks	**5 Marks**	**6 – 7 Marks**	**8 – 10 Marks**	
Feature: Offline mode **[10 Marks]**	• Feature not included. • Feature implemented but very buggy.	• Feature is working mostly with some bugs.	• Feature is working with only minor bugs.	• Feature excellently implemented.	
	0 – 4 Marks	**5 Marks**	**6 – 7 Marks**	**8 – 10 Marks**	
Feature: Real-time Notification **[10 Marks]**	• Feature not included. • Feature implemented but very buggy.	• Feature is working mostly with some bugs.	• Feature is working with only minor bugs.	• Feature excellently implemented.	
	0 – 4 Marks	**5 Marks**	**6 – 7 Marks**	**8 – 10 Marks**	

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PORTFOLIO OF EVIDENCE (POE)					
Marking Criteria	**Does not meet the required standard**	**Meets the required standard**	**Partially exceeds the required standard**	**Greatly exceeds the required standard**	**Feedback**
Feature: Multi-language	• Feature not included. • Feature implemented but very buggy.	• Feature is working mostly with some bugs.	• Feature is working with only minor bugs.	• Feature excellently implemented.	
[10 Marks]	**0 – 4 Marks**	**5 Marks**	**6 – 7 Marks**	**8 – 10 Marks**	
Feature: User Defined 4	• Feature not included. • Feature implemented but very buggy.	• Feature is working mostly with some bugs.	• Feature is working with only minor bugs.	• Feature excellently implemented.	
[10 Marks]	**0 – 4 Marks**	**5 Marks**	**6 – 7 Marks**	**8 – 10 Marks**	
Feature: User Defined 5	• Feature not included. • Feature implemented but very buggy.	• Feature is working mostly with some bugs.	• Feature is working with only minor bugs.	• Feature excellently implemented.	
[10 Marks]	**0 – 4 Marks**	**5 Marks**	**6 – 7 Marks**	**8 – 10 Marks**	
The app has been prepared for publication in the Play Store.	• No preparation has been done. • No evidence of preparation has been submitted. • The app is not ready for publication.	• The minimum requirements for Play Store publication are met.	• The minimum requirements for Play Store publication are met. • Some additional material is included.	• The app is completely ready to be published to the Play Store. • All material, including screenshots for the Play Store, is included.	
[5 Marks]	**0 – 1 Marks**	**2 – 3 Marks**	**4 Marks**	**5 Marks**	

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AI-use writeup **[5 Marks]**	• Writeup not included.	• Writeup included, but the details provided are vague.	• Writeup does not include details of how AI was used and cited.	• A detailed write-up which includes information on how it was used and cited. • Mark may be awarded if AI is not used at all.	
	0 – 1 Marks	**2 – 3 Marks**	**4 Marks**	**5 Marks**	
User interface **[10 Marks]**	• Not user-friendly, confusing, and illogical. • Somewhat user-friendly with lots of mistakes in the design.	• Mostly user-friendly, with some mistakes in the design. • Somewhat consistent use of layout, fonts, and colours.	• Mostly user-friendly, with some attempt made at making the capture process easier for the user. • Mostly consistent use of layout, fonts, and colours.	• Excellent and user-friendly design. • Completely consistent use of layout, fonts, and colours.	
	0 – 4 Marks	**5 Marks**	**6 – 7 Marks**	**8 – 10 Marks**	
GitHub, Readme and automated testing **[10 Marks]**	• Zip file added or file upload used for GitHub.	• Multiple commits and unit testing are not working. • No readme file.	• Automated testing includes, but is not working correctly • Somewhat of a readme file created.	• Readme file created and implemented correctly. • Automated testing is in place. • Multiple commits.	
	0 – 4 Marks	**5 Marks**	**6 – 7 Marks**	**8 – 10 Marks**	

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Demonstration video **[5 Marks]**	• No demonstration video included. • Informal and unprepared, or not showing all the features or no voice-over included.	• It is not completely professional, but all features are demonstrated.	• Mostly professional with some issues, but all features were demonstrated.	• Fully professional video showing all the required features in detail.	
	0 – 1 Marks	**2 – 3 Marks**	**4 Marks**	**5 Marks**	**PoE Total /100**

[TOTAL MARKS: 300]

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